

Math 580 Assignment 5

Edward Chernysh

Problem 1

Let φ be a non-trivial smooth function in $\mathbb{R}^n$ having compact support inside B . For a given $k \in \mathbb{N}$, we define a new function φ_k according to the rule:

$$\varphi_k : \mathbb{R}^n \rightarrow \mathbb{R}, \quad \varphi_k(x) := k^{\frac{n-2}{2}} \varphi(kx).$$

Since $\text{supp}(\varphi) \subset B$, it is not hard to see that each φ_k is smooth with compact support in B . Thus, $\{\varphi_k\}_k$ is a sequence in $H^1(B)$. Letting $1 \leq p < \infty$ be given, we compute

$$\begin{aligned} \|\varphi_k\|_{L^p(B)}^p &= \int_{\mathbb{R}^n} |\varphi_k|^p = \int_{\mathbb{R}^n} k^{\frac{p(n-2)}{2}} |\varphi(kx)|^p \, dx \\ &= k^{\frac{p(n-2)}{2}} \int_{\mathbb{R}^n} |\varphi(kx)|^p \, dx \\ &= k^{\frac{p(n-2)}{2}} \cdot k^{-n} \int_{\mathbb{R}^n} |\varphi(x)|^p \, dx \\ &= k^{\frac{p(n-2)}{2}} \cdot k^{-n} \|\varphi\|_{L^p(B)}^p. \end{aligned}$$

Similarly,

$$\begin{aligned} \|\nabla \varphi_k\|_{L^p(B)}^p &= \int_{\mathbb{R}^n} |\nabla \varphi_k|^p = k^{\frac{p(n-2)}{2}} \int_{\mathbb{R}^n} |\nabla_x \varphi(kx)|^p \, dx \\ &= k^{\frac{p(n-2)}{2}} \int_{\mathbb{R}^n} k^p |\nabla \varphi(kx)|^p \, dx \\ &= k^{\frac{p(n-2)}{2} + p} k^{-n} \|\nabla \varphi\|_{L^p(B)}^p \\ &= k^{\frac{pn}{2}} k^{-n} \|\nabla \varphi\|_{L^p(B)}^p. \end{aligned}$$

Taking $p = 2$, we see that

$$\|\varphi_k\|_{L^2(B)}^2 = k^{n-2} \cdot k^{-n} \|\varphi\|_{L^2(B)}^2 = \frac{1}{k^2} \|\varphi\|_{L^2(B)}^2$$

which is bounded uniformly in k . Similarly,

$$\|\nabla \varphi_k\|_{L^2(B)}^2 = \|\nabla \varphi\|_{L^2(B)}^2$$

which does not depend on k . Hence, $\{\varphi_k\}$ is a bounded sequence in $H^1(B)$. On the other hand, taking $p = q = 2^*$, the above tells us that

$$\|\varphi_k\|_{L^q(B)}^q = k^{\frac{2n}{n-2} \cdot \frac{(n-2)}{2}} \cdot k^{-n} \|\varphi\|_{L^q(B)}^q = \|\varphi\|_{L^q(B)}^q \neq 0.$$

By way of contradiction, suppose that $\{\varphi_k\}$ has a subsequence in $L^q(B)$ that converges to some $f \in L^q(B)$. Let us keep the notation $\{\varphi_k\}$ for this subsequence. After passing to a subsequence, we might as well assume that φ_k converges pointwise almost everywhere to f on B . However, φ_k converges pointwise almost everywhere to 0. Hence, we must have $f = 0$ almost everywhere on B , which directly contradicts the fact that

$$\|f\|_{L^q(B)}^q = \lim_{k \rightarrow \infty} \|\varphi_k\|_{L^q(B)}^q = \|\varphi\|_{L^q(B)}^q \neq 0.$$

Problem 2

We divide this problem into two parts. By class results, it is enough to check that S is totally bounded in $L^p(\Omega)$. Let now $r > 0$ be given and take $\varepsilon > 0$ small (this is to be made precise later). By hypothesis, we can find $j \gg 1$ with the property that

$$\|u\|_{L^p(\Omega - K_j)} < \varepsilon, \quad \forall u \in S.$$

Thus, for $\delta > 0$ small enough, we have

$$\|\Delta_h u\|_{L^p(\Omega_h)} \leq |h| \|\nabla u\|_{L^p(\Omega)} \leq M |h| < \varepsilon$$

with $h \in B_\delta(0)$ and $u \in S$. Here, $M > 0$ is a fixed uniform bound on the $W^{1,p}$ -norms of the family S . Henceforth, we will assume that we have taken j large and $\delta > 0$ small.

As in the notes, define

$$\lambda := \frac{\delta}{2n}$$

and put

$$G_\lambda := \{\lambda a + [0, \lambda]^n : a \in \mathbb{Z}^n\}.$$

Since each K_j is compact, there are only finitely many cubes $Q_1, \dots, Q_m$ from G_λ having non-empty intersection with K_j . For such cubes, we define $\Sigma := \bigcup_{l=1}^m Q_l$. Making $\delta > 0$ smaller if necessary, we may suppose that $\Sigma \subseteq \Omega$ and $\text{dist}(\Sigma, \partial\Omega) > \delta$.

We now define $X = \text{span}\{Q_i : 1 \leq i \leq m\}$. Since X is finite dimensional, the Heine-Borel property holds. In particular, the ball

$$X_M = \{\phi \in X : \|\phi\|_{L^p} \leq M\}$$

is totally bounded. Thus, we can find finitely many open balls of radius $r/2$ in $L^2(\Omega)$ that cover X_M . It remains to show that if we enlarge these balls to have radius r , then we obtain a cover for the set S . Put otherwise, we will establish that given $f \in S$ one can find $g \in X_M$ such that $\|f - g\|_{L^p} < r/2$. Once we have shown this, we will have the desired covering by balls.

To this end, fix $f \in S$ and define

$$g = \sum_{i=1}^m \left[\frac{1}{m(Q_i)} \int_{Q_i} f \right] \chi_{Q_i}$$

By means of Hölder's inequality we obtain

$$\begin{aligned} \|g\|_{L^p} &= \sum_{i=1}^m \left[\frac{1}{m(Q_i)} \int_{Q_i} f \right]^p m(Q_i) = \sum_{i=1}^m \frac{1}{m(Q_i)^{p-1}} \left[\int_{Q_i} f \right]^p \\ &\leq \sum_{i=1}^m \int_{Q_i} |f|^p \\ &\leq M^p. \end{aligned}$$

Taking the p^{th} root on either side we see that $\|g\|_{L^p} \leq M$ whence $g \in X_M$.

Now, notice that

$$\begin{aligned} \|f - g\|_{L^p(\Omega)} &\leq \|f - g\|_{L^p(\Omega \setminus \Sigma)} + \|f - g\|_{L^p(\Sigma)} = \|f\|_{L^p(\Omega \setminus \Sigma)} + \|f - g\|_{L^p(\Sigma)} \\ &\leq \varepsilon + \|f - g\|_{L^p(\Sigma)} \end{aligned}$$

where the last inequality follows from $\|f\|_{L^p(\Omega \setminus \Sigma)} \leq \|f\|_{L^p(\Omega - K_j)} < \varepsilon$. In hopes of bounding the second term, we once again invoke Hölder's inequality;

$$\begin{aligned} \|f - g\|_{L^p(\Sigma)}^p &= \sum_{i=1}^m \int_{Q_i} \left| f(x) - \frac{1}{m(Q_i)} \int_{Q_i} f \right|^p dx \\ &= \sum_{i=1}^m \int_{Q_i} \left| \frac{1}{m(Q_i)} \int_{Q_i} (f(x) - f(y)) dy \right|^p dx \\ &= \sum_{i=1}^m \frac{1}{m(Q_i)} \int_{Q_i} \int_{Q_i} |f(x) - f(y)|^p dy dx. \end{aligned}$$

Recalling our choice of constants δ and λ , we see that the above may be bounded further

$$\begin{aligned} \|f - g\|_{L^p(\Sigma)}^p &\leq \sum_{i=1}^m \frac{1}{m(Q_i)} \int_{Q_i} \int_{B_\delta(x)} |f(x) - f(y)|^p dy dx \\ &= \frac{1}{\lambda^n} \sum_{i=1}^m \int_{Q_i} \int_{B_\delta} |f(x) - f(x+h)|^p dh dx \\ &= \frac{1}{\lambda^n} \int_{B_\delta} \|\Delta_h f\|_{L^p(\Sigma)}^p dh \leq \frac{1}{\lambda^n} m(B_\delta) \varepsilon^p = \omega_n (2n)^n \varepsilon^p \end{aligned}$$

Combining the above inequalities we see that, for a suitably small choice of $\varepsilon > 0$ there holds $\|g\|_{L^p} < r/2$ which concludes the proof.

For the second part, let us consider the domain

$$\Omega := \{(x, y) \in \mathbb{R}^2 : 0 < y < e^{-x^2}\}.$$

Clearly, Ω is unbounded. Let us fix a bounded subset S of $H_0^1(\Omega)$; it is enough to check that it satisfies the uniform decay condition. To this end, let $M > 0$ be such that $\|u\|_{H^1(\Omega)} \leq M$ for all $u \in S$. Since Ω has finite measure and $\text{dist}(x, \partial\Omega)$ tends to 0 as x tends to $\pm\infty$, it suffices to establish a bound as in equation (52) of Theorem 10 in the notes. Since we have a smooth boundary, the procedure for showing this is identical to the notes. For the sake of completeness, we explain the procedure.

The boundary of Ω is given by the following union of sets:

$$\partial\Omega = \{(x, 0) : x \in \mathbb{R}\} \sqcup \{(x, y) : x \in \mathbb{R}, y = e^{-x^2}\}.$$

We will prove that the desired bound holds for all open sets near “on” the part of the boundary where $y = 0$. After straightening out the boundary, we may repeat the process on the section $y = e^{-x^2}$.

Given $h > 0$, consider first an arbitrary test function ϕ satisfying $\|\phi\|_{H^1} \leq M$. Note that we may extend ϕ to be 0 outside of Ω . For any $\delta > 0$, define $\Sigma_\delta = (x_0, x_0 + h) \times (0, \delta)$ for some arbitrary $x_0 \in \mathbb{R}$. Now, notice that for any $(x, y) \in \Sigma_\delta$ there holds

$$\phi(x, y) = \int_0^y \partial_y \phi(x, t) dt$$

for some $0 \leq t \leq h$. By Hölder’s inequality,

$$|\phi(x, y)|^2 \leq h \int_0^h |\partial_y \phi(x, t)|^2 dt$$

Integrating either side over $y \in (0, \delta)$ we see that

$$\int_0^\delta |\phi(x, y)|^2 dy \leq h\delta \int_0^h |\partial_y \phi(x, t)|^2 dt.$$

Finally, integrating over $x \in (x_0, x_0 + h)$ yields

$$\int_{\Sigma_\delta} |\phi(x, y)|^2 dy \leq h\delta \int_{\Sigma_h} |\partial_y \phi(x, t)|^2 dt.$$

Combining the above inequality with $\|\phi\|_{H^1} \leq M$ we see that $\|\phi\|_{L^2(\Sigma_\delta)} \leq C\sqrt{\delta}$ for some constant $C > 0$. By the density of test function in $H_0^1(\Omega)$, we conclude that $\|u\|_{L^2(\Sigma_\delta)} \leq C\sqrt{\delta}$ for all $u \in S$.

We have thus obtained the desired bound and, as in class, the result readily follows.

Problem 3

- (a) It is indeed possible to have $C = \infty$ on an unbounded domain. The example we now construct is due to J.A. Hummel's useful paper¹. Henceforth, we identify $\mathbb{R}^2$ with $\mathbb{C}$. Consider now the bounded domain described by the set

$$\Omega := \{(e^{-y} \cos(x), e^{-y} \sin(x)) \mid x > 1, (x + 2\pi)^{-2} < y < x^{-2}\}.$$

Then, Ω is in biholomorphic equivalence with the set

$$\{z = x + iy \in \mathbb{C} : x > 1, (x + 2\pi)^{-2} < y < x^{-2}\}.$$

Namely, there is a biholomorphism between these two sets (take $z \mapsto e^{iz}$ for instance). We define a function $f : \Omega \rightarrow \mathbb{R}$ by the rule

$$f(e^{-y} \cos(x), e^{-y} \sin(x)) = x - \alpha$$

with $\alpha \in \mathbb{R}$ such that $\int_{\Omega} f = 0$. Note that such an α exists by the intermediate value theorem. After a change of variables in $\mathbb{R}^2$, we can see that $\|f\|_{L^2(\Omega)} = \infty$ and that $\nabla u \in L^2(\Omega)$. However, f will be in L^2_{loc} and therefore we can, by following constructions from the notes, find a sequence in $H^1(\Omega)$ that converges to f locally uniformly on Ω , whence we see that the inequality cannot hold for $C < \infty$.

- (b) Let $A : D_A \rightarrow L^2(\Omega)$ be the extension of the Laplace operator $-\Delta$ discussed in the notes. We know that this operator has eigenvalues

$$\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_k \leq \dots$$

with corresponding eigenvectors $\{\varphi_k\}$. In fact, since constant functions are eigenvectors with $\lambda = 0$, we see that $\lambda_1 = 0$. Note also that $u \in H^1(\Omega)$ being an eigenvector for $\lambda = 0$ means that $\int_{\Omega} |\nabla u|^2 = 0$. Hence, any eigenvector for λ_1 will be a constant multiple of the unit vector

$$u_1 := a.$$

for some constant $a \neq 0$. Put otherwise, λ_1 has multiplicity 1 and so u_2 does not correspond to 0. In particular, $\lambda_2 > 0$. Recall that there exists $C > 0$ for which

$$\|u\|_{L^2(\Omega)}^2 \leq C \|\nabla u\|^2, \quad \forall u \in \text{span}(u_1)^\perp.$$

In fact, we know that the optimal $C > 0$ for which the above holds is given by

$$C := \frac{1}{\lambda_2}.$$

¹Titled "Counter Examples to the Poincaré inequality."

Fixing $u \in H^1(\Omega)$, we calculate

$$\begin{aligned}\langle u - \bar{u}, u_1 \rangle_{L^2} &= a \int_{\Omega} (u - \bar{u}) = a \int_{\Omega} u - a \int_{\Omega} \bar{u} \\ &= a \int_{\Omega} u - a \int_{\Omega} u = 0.\end{aligned}$$

Hence, we will have $u - \bar{u} \in \text{span}(u_1)^\perp$ so that

$$\|u - \bar{u}\|_{L^2(\Omega)}^2 \leq \lambda_2^{-1} \|\nabla(u - \bar{u})\|^2 = \lambda_2^{-1} \|\nabla u\|^2.$$

Thus, if C_* denotes the optimal constant in (*), we must have $C_* \leq \lambda_2^{-1}$. It only remains to show that $C_* \geq \lambda_2^{-1}$. To achieve this, we will show that equality is achieved in (*) for some function $u \in H^1(\Omega)$ and $C := \lambda_2^{-1}$. Let u_2 be the eigenvector associated to $\lambda_2 \neq \lambda_1$. Then,

$$\int_{\Omega} u_2 = \frac{1}{a \cdot m(\Omega)} \int_{\Omega} u_2 u_1 = \frac{1}{a \cdot m(\Omega)} \langle u_1, u_2 \rangle_{L^2} = 0.$$

It follows that

$$\|u_2 - \bar{u}_2\|_{L^2}^2 = \|u_2\|_{L^2(\Omega)}^2.$$

On the other hand, we know that

$$\int_{\Omega} |\nabla u_2|^2 = \lambda_2 \int_{\Omega} u_2^2.$$

Putting these together yields

$$\|u_2 - \bar{u}\|_{L^2}^2 = \|u_2\|_{L^2(\Omega)}^2 = \lambda_2^{-2} \|\nabla u_2\|_{L^2(\Omega)}^2.$$

Thus, we must have $C_* \geq \lambda_2^{-2}$ and equality follows. We deduce that the best constant C for which equation (*) holds true is λ^{-1} where λ is the smallest non-zero eigenvalue of A .

- (c) Consider the domain $\Omega = (0, a) \times (0, b)$ for some $a, b > 0$. For a pair (n, m) with $n, m \geq 0$, we define

$$\lambda_{n,m} := \left(\frac{\pi n}{a}\right)^2 + \left(\frac{\pi m}{b}\right)^2$$

and

$$u_{n,m}(x, y) = \cos\left(\frac{\pi n x}{a}\right) \cos\left(\frac{\pi m y}{b}\right).$$

It is well known that these functions form an orthonormal basis for $L^2(\Omega)$ (this is usually an exercise in analysis 4). To see this more explicitly, we first note that

$$-\Delta u_{n,m} \equiv \lambda_{n,m} u_{n,m}$$

for all $n, m \in \mathbb{N}_0$. Indeed, this is by direct calculation. To see that these form a basis for $L^2(\Omega)$, simply note that the individual functions

$$\cos\left(\frac{\pi nx}{a}\right)$$

form a basis for $L^2((0, a))$ (and likewise for $(0, b)$). Thus, the claim follows by Fubini's theorem. In any case, using this, we will be able to compute the best constant $C > 0$ for which the given inequality holds.

After a change of coordinates, we may assume that $b \leq a$. Since 0 is an eigenvalue of $-\Delta$, the second eigenvalue will be given by

$$\lambda_{1,0} = \frac{\pi^2}{a^2}.$$

Hence, the optimal constant for (*) is explicitly given by

$$C = \frac{1}{\lambda_{1,0}}.$$

Then, the corresponding eigenvector $u := u_{1,0}$ must achieve this $C > 0$ in (*). Indeed, being an eigenvector of the operator A in the notes means that

$$\int_{\Omega} |\nabla u|^2 = \frac{1}{C} \int_{\Omega} |u|^2.$$

On the other hand, one can check that $\bar{u} = 0$ by direct computation. Thus,

$$\frac{\int_{\Omega} |u - \bar{u}^2|}{\int_{\Omega} |\nabla u|^2} = C.$$

Hence, the constant C is achieved.

Problem 4

We seek to characterize all functions $u \in H_0^1(\Omega)$ which satisfy the equation

$$-\Delta u = \lambda u \quad \text{in } \Omega,$$

for $\lambda \in \mathbb{R}$. To achieve this, we will first seek out *smooth* solutions to this eigenvalue problem. As it turns out, these will form a complete system in $L^2(\Omega)$. First, recall that in the case of $H_0^1(\Omega)$, all the eigenvalues of the Laplacian will be positive.

As a first step, we switch to polar coordinates whence the eigenvalue problem can be written as follows:

$$\Delta u = \frac{1}{r} \partial_r u + \partial_r^2 u + \frac{1}{r^2} \partial_{\theta}^2 u = \lambda u.$$

We will solve this problem by way of the *separation of variables method*. Assume that $u(r, \theta) = f(r)g(\theta)$, for some functions f and g of a single variable. If in addition g is assumed to have the form

$$g(\theta) = a \sin(n\theta) + b \cos(n\theta)$$

for $n \in \mathbb{N}$ and $a, b \in \mathbb{R}$, then g will satisfy the ODE $g'' = -n^2 g^2$. With this in mind, we calculate

$$\Delta u = \frac{gf'}{r} + gf'' + \frac{fg''}{r^2} = g \left(\frac{f'}{r} + f'' - \frac{n^2}{r^2} f \right).$$

Hence, the equation $\Delta u + \lambda u = 0$ can be solved by finding a solution to

$$0 = \frac{f'}{r} + f'' - \frac{n^2}{r^2} f + \lambda u = 0. \quad (1)$$

However, solutions to the equation above are the Bessel functions, which are often denoted via the following:

$$J_n(r) := \sum_{j \in \mathbb{N}_0} \frac{(-1)^j}{j!(n+j)!} \left(\frac{r}{n} \right)^{2+2j}.$$

As we are looking for solutions having the Dirichlet boundary condition, we should rescale the function above to something that vanishes on the boundary of the disk D_R . Let now $\{\zeta_{n,j}\}_{j \in \mathbb{N}_0}$ be an enumeration of the zeros of $J_n(r)$, for a fixed n . Put then

$$f_{n,j}(r) := J_n \left(\frac{\zeta_{n,j}}{R} r \right)$$

which clearly vanishes on the boundary of the disk. By our calculations the product $f_{n,j}g$ will solve the differential equation stated above for a certain λ . More precisely, $f_{n,j}g$ will satisfy the equation

$$-\Delta f_{n,j}g \equiv \left(\frac{\zeta_{n,j}^2}{R^2} \right) f_{n,j}g \quad \text{in } \Omega$$

with Dirichlet boundary conditions. Hence, we can write solutions to the eigenvalue problem on the disk using the following functions:

$$a_{n,j} := J_n \left(\frac{\zeta_{n,j}}{R} r \right) \cos(n\theta) \quad \text{and} \quad b_{n,j} := J_n \left(\frac{\zeta_{n,j}}{R} r \right) \sin(n\theta).$$

By following the calculations in Problem 3 (together with messy identities for the Bessel & trigonometric functions), this family can be shown to be complete in $L^2(B_R)$

Problem 5

We first setup some heavy machinery. Using that ρ is smooth and positive on $\bar{\Omega}$, we can find $\alpha_1, \alpha_2 > 0$ such that

$$\alpha_1 \leq \rho(x) \leq \alpha_2$$

for all $x \in \bar{\Omega}$. Next, let $t > 0$ be sufficiently large (how large will be determined later). Consider the operator

$$J : L^2(\Omega) \rightarrow (H_0^1(\Omega))^*$$

where we define Jf according to the rule

$$(Jf)(v) = \int_{\Omega} \rho f v, \quad v \in H_0^1(\Omega).$$

Since $f \in L^2(\Omega)$, it is obvious that $Jf \in (H_0^1(\Omega))^*$ and that J itself a well defined linear operator. Suppose that $Jf = 0$ for some $f \in L^2(\Omega)$. This is to say that $\int_{\Omega} \rho f v = 0$ for all $v \in H_0^1(\Omega)$. In particular, this holds for all test functions $v \in C_c^\infty(\Omega)$. Therefore, we must have $\rho f = 0$ almost everywhere on Ω whence we find that $f = 0$ in $L^2(\Omega)$. This means that J is injective, and hence is an embedding

$$J : L^2(\Omega) \hookrightarrow (H_0^1(\Omega))^*.$$

Now, let $\tilde{A} : H_0^1(\Omega) \rightarrow (H_0^1(\Omega))^*$ be the operator given by

$$(\tilde{A}u)(v) = a(u, v) = \int_{\Omega} \left(\sum_{i,j=1}^n a_{ij} \partial_i u \partial_j v \right) + \int_{\Omega} b u v. \quad (2)$$

Next, we consider the family

$$D_A := \{u \in H_0^1(\Omega) : \tilde{A}u \in \text{Im}(J)\}.$$

Note that D_A is non-empty as it contains the zero function. We define a mapping $A : D_A \rightarrow L^2(\Omega)$ via the composition

$$A : D_A \xrightarrow{\tilde{A}} \text{Im}(J) \xrightarrow{J^{-1}} L^2(\Omega), \quad A := J^{-1} \circ \tilde{A}.$$

As the composition of linear operators, A is itself linear. Next, we consider for $t > 0$ sufficiently large the operator

$$A + tI : D_A \rightarrow L^2(\Omega),$$

where I is the identity operator $L^2(\Omega) \rightarrow L^2(\Omega)$. Evidently, $A + tI$ is linear. To see that it is injective, suppose that

$$(A + tI)(u) = 0$$

for some $u \in D_A \subseteq H_0^1(\Omega)$, where the equality above takes place in $L^2(\Omega)$. Applying the operator J to both sides, we see that

$$\tilde{A}u + tJ(u) = 0,$$

where this equality holds in the $(H_0^1(\Omega))^*$ sense. That is, we have

$$\int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i u \partial_j v + \int_{\Omega} b u v + t \int_{\Omega} \rho u v = 0, \quad \forall v \in H_0^1(\Omega).$$

Taking $v := u$, we discover that

$$\int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i u \partial_j u + \int_{\Omega} (b + t\rho) u^2 = 0.$$

Using the ellipticity condition, this implies that

$$0 \geq \int_{\Omega} \alpha |\nabla u|^2 + \int_{\Omega} (b + t\rho) u^2 \geq \int_{\Omega} \alpha |\nabla u|^2 + \int_{\Omega} (b + t\alpha_1) u^2.$$

Thus, by taking $t > 0$ so large that $b + t\alpha_1 \geq \alpha$ in $\bar{\Omega}$, the above tells us that

$$0 \geq \int_{\Omega} \alpha |\nabla u|^2 + \int_{\Omega} (b + t\alpha_1) u^2 \geq \alpha \|u\|_{H^1}^2.$$

Note that this very same argument shows that the (bounded) bilinear map

$$H_0^1(\Omega) \times H_0^1(\Omega) \ni (u, v) \mapsto \int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i u \partial_j v + \int_{\Omega} b u v + t \int_{\Omega} \rho u v$$

is an inner product inducing a norm equivalent to that of $H_0^1(\Omega)$, whenever t is sufficiently large. Hence, we must have $u = 0$ whence $A + tI$ is injective for all t large, say, for all $t > t_0$. Next, we check that $A + tI$ is surjective. For this, we will make use of the Riesz representation theorem. Given $f \in L^2(\Omega)$, we know that Jf belongs to the topological dual of $H_0^1(\Omega)$. Hence, by the Riesz representation theorem (and the fact that the bilinear map above is an inner product), we can find $u \in H_0^1(\Omega)$ with the property that

$$\int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i u \partial_j v + \int_{\Omega} b u v + t \int_{\Omega} \rho u v = (Jf)(v) = \int_{\Omega} \rho f v$$

for all $v \in H_0^1(\Omega)$. Note that this is the same as writing

$$\tilde{A}u + tJu = Jf \in \text{Im}(J), \tag{3}$$

with the equality taking place in $(H_0^1(\Omega))^*$. Since J is injective, we can apply J^{-1} to both sides of this equation, and in doing so we obtain

$$Au + tIu = (A + tI)u = f, \quad \text{in } L^2(\Omega).$$

It remains to check that $u \in D_A$, but this will follow quite easily from the equality in (3). Indeed, since $tJu \in \text{Im}(J)$, we must have $\tilde{A}u \in \text{Im}(J)$ as well. We conclude that $A + tI$ is bijective when viewed as an operator $D_A \rightarrow L^2(\Omega)$ (for all $t > t_0$). Next, we fix $t > t_0$ and consider the resolvent

$$R_t : L^2(\Omega) \rightarrow D_A \subseteq H_0^1(\Omega) \hookrightarrow L^2(\Omega)$$

where the latter embedding is compact (by the Rellich-Kondrachov theorem this is very mild assumption – one would only need $\partial\Omega$ to be nice enough). Before proceeding further, we establish some properties of this functional R_t .

- We claim that R_t is a bounded operator $L^2(\Omega) \rightarrow H_0^1(\Omega)$. To see this, let $f \in L^2(\Omega)$ be given and put $u := R_t f \in D_A$. That is, we have

$$u = (A + tI)^{-1}f \implies (Au + tu) = f$$

in $L^2(\Omega)$. Applying the operator J to both sides, we find that

$$\int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i u \partial_j v + \int_{\Omega} buv + t \int_{\Omega} \rho uv = \int_{\Omega} \rho f v, \quad \forall v \in H_0^1(\Omega).$$

Taking $v := u$ in the above and applying the uniform ellipticity condition once more, we obtain (since $t > t_0$)

$$\int_{\Omega} \rho f u \geq \int_{\Omega} \alpha |\nabla u|^2 + \int_{\Omega} (b + t\rho) u^2 \geq \alpha \|u\|_{H^1}^2.$$

Hölder's inequality also gives

$$\alpha \|u\|_{H^1}^2 \leq \int_{\Omega} \rho f u \leq C \|f\|_{L^2(\Omega)} \|u\|_{H^1(\Omega)},$$

where C is a bound on $|\rho|$ over $\bar{\Omega}$. Clearly, the inequality above shows that

$$\|R_t f\|_{H^1(\Omega)} = \|u\|_{H^1(\Omega)} \leq C \|f\|_{L^2(\Omega)}$$

for all $f \in L^2(\Omega)$. Hence, R_t is bounded.

- By the Rellich-Kondrachov theorem, R_t will be compact when viewed as a map

$$L^2(\Omega) \rightarrow L^2(\Omega).$$

Let us now endow $L^2(\Omega)$ with an alternative (but equivalent) inner product. Define a bilinear form on $L^2(\Omega)$ by the rule

$$\langle f, g \rangle_p := \int_{\Omega} \rho f g.$$

Since ρ is bounded on Ω , Hölder's inequality immediately shows that this map is bounded. Symmetry and linearity are clear as well. Note also that

$$0 \leq \langle f, f \rangle_p = \int_{\Omega} \rho f^2 \leq C \|f\|_{L^2(\Omega)}^2$$

for a constant $C > 0$ depending only on ρ . Similarly, since ρ is strictly positive on $\bar{\Omega}$, for $\alpha_1 > 0$ we have

$$\langle f, f \rangle_p = \int_{\Omega} \rho f^2 \geq \alpha_1 \|f\|_{L^2(\Omega)}^2.$$

Thus, $\langle f, f \rangle_p$ is equivalent to the standard inner product on $L^2(\Omega)$. We claim that, with respect to this inner product, R_t is both positive and symmetric. First, we establish the positivity. Given $f \in L^2(\Omega)$, let $u := R_t f$ so that

$$(A + tI)u = Au + tu = f \quad \text{in } L^2(\Omega).$$

Applying the operator J to both sides, we discover that

$$\int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i u \partial_j v + \int_{\Omega} b u v + t \int_{\Omega} \rho u v = \int_{\Omega} \rho f v, \quad \forall v \in H_0^1(\Omega).$$

With $v := u$, the above yields

$$\begin{aligned} \langle f, f \rangle_p &= \int_{\Omega} \rho f^2 = \int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i u \partial_j u + \int_{\Omega} (b + t\rho) u^2 \\ &\geq \alpha \int_{\Omega} |\nabla u|^2 + \int_{\Omega} (b + t\rho) u^2 \\ &\geq \alpha \|u\|_{H^1(\Omega)}^2. \end{aligned}$$

Hence, R_t is positive with respect to $\langle \cdot, \cdot \rangle_p$. As for symmetry, for $f, g \in L^2(\Omega)$ we define $\bar{f} := R_t f$ and $\bar{g} := R_t g$. Arguing as above, we have the identities

$$\int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i \bar{f} \partial_j v + \int_{\Omega} b \bar{f} v + t \int_{\Omega} \rho \bar{f} v = \int_{\Omega} \rho f v, \quad \forall v \in H_0^1(\Omega), \quad (4)$$

$$\int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i \bar{g} \partial_j v + \int_{\Omega} b \bar{g} v + t \int_{\Omega} \rho \bar{g} v = \int_{\Omega} \rho g v, \quad \forall v \in H_0^1(\Omega). \quad (5)$$

In the first equation, we set $v := \bar{g}$ whilst in the second we take $v := \bar{f}$, thereby obtaining:

$$\int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i \bar{f} \partial_j \bar{g} + \int_{\Omega} b \bar{f} \bar{g} + t \int_{\Omega} \rho \bar{f} \bar{g} = \int_{\Omega} \rho f \bar{g}, \quad (6)$$

$$\int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i \bar{g} \partial_j \bar{f} + \int_{\Omega} b \bar{g} \bar{f} + t \int_{\Omega} \rho \bar{g} \bar{f} = \int_{\Omega} \rho g \bar{f}. \quad (7)$$

Combining these, we see that

$$\int_{\Omega} \rho \bar{f} g = \int_{\Omega} \rho \bar{g} f.$$

That is, $\langle \bar{f}, g \rangle_p = \langle f, \bar{g} \rangle_p$ whence R_t is symmetric. To summarize this procedure, we have shown that R_t is a positive compact symmetric operator $L^2(\Omega) \rightarrow L^2(\Omega)$. Thus, we can apply the spectral theorem to obtain a complete family of eigenvectors $\{\varphi_k\}_k$ with countably many corresponding eigenvalues $\{\mu_k\}_k$ in $\mathbb{R}$. By this same theorem, every eigenvalue will have finite multiplicity. Next, we relate the eigenvalues/eigenvectors of R_t to those of A .

First, let $\varphi \in L^2(\Omega)$ be an eigenvalue of R_t with corresponding eigenvalue μ . Then, we have

$$(A + tI)^{-1} \varphi = \mu \varphi \implies \varphi = \mu(A + tI) \varphi = \mu A(\varphi) + t\mu \varphi.$$

Since R_t is positive, we have $\mu > 0$ so that

$$A\varphi = \left(\frac{1}{\mu} - t\right) \varphi. \quad (8)$$

Conversely, if φ is an eigenvector of A with eigenvalue λ , then, $A\varphi = \lambda\varphi$ forces

$$(A + tI)(\varphi) = A\varphi + t\varphi = (\lambda + t)\varphi$$

whence $\varphi = (\lambda + t)R_t\varphi$ and φ is an eigenvector of R_t with eigenvalue

$$\mu = \frac{1}{\lambda + t}.$$

Note also that the formula (8) yields a correspondence between the eigenvalues of R_t and those of A . Since the spectral theorem guarantees that the eigenvalues of R_t are countable and real, the same can be said for those of A . Furthermore, since the eigenvalues $\{\mu_k\}$ of R_t converge to 0 as $k \rightarrow \infty$, (3) shows that the eigenvalues $\{\lambda_k\}$ of A satisfy $\lambda_k \rightarrow \infty$ as $k \rightarrow \infty$. This answers (a).

Next, we treat (b). Again, by the spectral theorem for positive and compact operators, we know that the eigenvectors of R_t form a complete orthonormal system in $L^2(\Omega)$ with respect to the inner product $\langle f, g \rangle_p$. By the correspondence between the eigenvectors of R_t and those of A , we are done with this part.

Let us now handle (c). Let $\{u_k\}$ be the eigenvectors of A obtained by applying the spectral theorem to R_t , as above. Then, $Au_k = \lambda_k u_k$ for each k , where $\lambda_k \in \mathbb{R}$ denotes the corresponding eigenvalue. Note that this last equality takes place in $L^2(\Omega)$. In particular, there holds

$$(A + tI)(u_k) = Au_k + tu_k = (\lambda_k + t)u_k.$$

Applying J to both sides, we find that

$$\tilde{A}u_k + tJ(u_k) = (\lambda_k + t)J(u_k),$$

the equality taking place in $(H_0^1(\Omega))^*$. That is, we have

$$\int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i u_k \partial_j v + \int_{\Omega} b u_k v + t \int_{\Omega} \rho u_k v = (\lambda_k + t) \int_{\Omega} \rho u_k v, \quad \forall v \in H_0^1(\Omega).$$

This can be rewritten as

$$a(u_k, v) + t \int_{\Omega} \rho u_k v = (\lambda_k + t) \langle u_k, v \rangle_p, \quad \forall v \in H_0^1(\Omega). \quad (9)$$

For all $t > t_0$, we know that the bilinear form on the left is an inner product on $H_0^1(\Omega)$. Since the $\{u_k\}$ are orthonormal in $L^2(\Omega)$ with respect to the inner product on the right hand side, we conclude that this family is orthogonal in $H_0^1(\Omega)$ with respect to the inner product on the left hand side. It remains to show that the family is also *complete* in $H_0^1(\Omega)$. However, this readily follows from the completeness of the system in $L^2(\Omega)$. Indeed, since every λ_k takes the form

$$\frac{1}{\mu_k} - t,$$

where $\mu_k > 0$ is an eigenvalue of R_t , we see that $(\lambda_k + t) > 0$ for all $k \in \mathbb{N}$. If we denote by $\langle u, v \rangle$ the inner product stated in (c), (9) tells us that

$$\langle u_k, v \rangle = (\lambda_k + t) \langle u_k, v \rangle_p$$

for all $k \in \mathbb{N}$. Thus, if $\langle u_k, v \rangle$ vanishes for all $k \in \mathbb{N}$, then $\langle u_k, v \rangle_p = 0$ for all k . Since the family $\{u_k\}$ forms a complete orthonormal system in $L^2(\Omega)$ with respect to $\langle f \rangle g_p$, we must have $v = 0$ in $L^2(\Omega)$, and also in $H_0^1(\Omega)$. This proves (c).

For (d), the claim follows from the regularity results established in the previous assignment for weak solutions to the same elliptic problem. Alternatively, we simply apply the elliptic regularity results from Evans.

DISCLAIMER: I discussed several problems with Dana Berman.